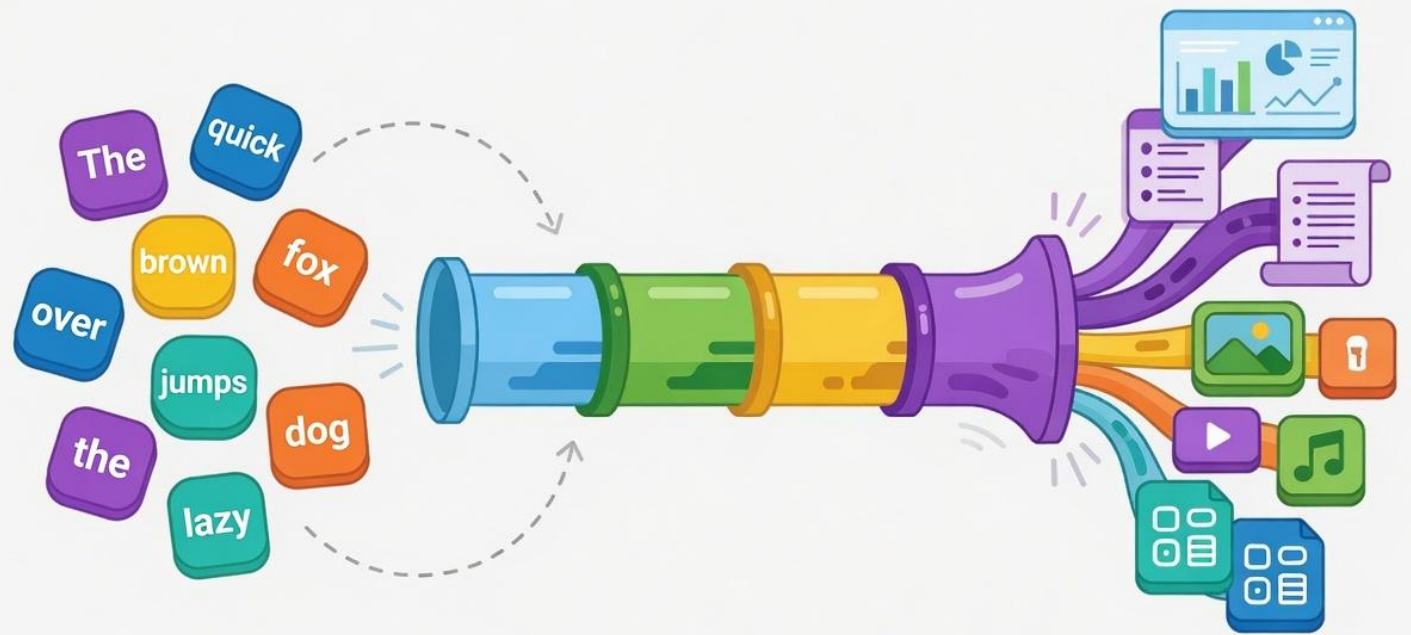


Token Economics



Agenda

- Introduction
- Demo 1 - 4 levers
- Demo 2 - Pricing
- Tracing and Scaling
- Resources

- Home
- My work
- Automations



@Maceta444 4 months ago

I hit my 1500 monthly credits just watching this video

14 Reply

0:02

0:29 / 4:59

Rory Preddy

10 Add robots.txt and sitemap.xml for search engines #5 enhancement seo
roryp/astro-coffee - roryp - 27m ago +13 -0

11 Add basic stylesheet for the landing page #1 enhancement css
roryp/astro-coffee - roryp - 28m ago +45 -0

12 Add MENU.md with current drinks and prices #2 enhancement content
roryp/astro-coffee - roryp - 28m ago +10 -0

⏸ ⏮ ⏭ ⚙ HD ⏪ ⏩ ↶ ↷

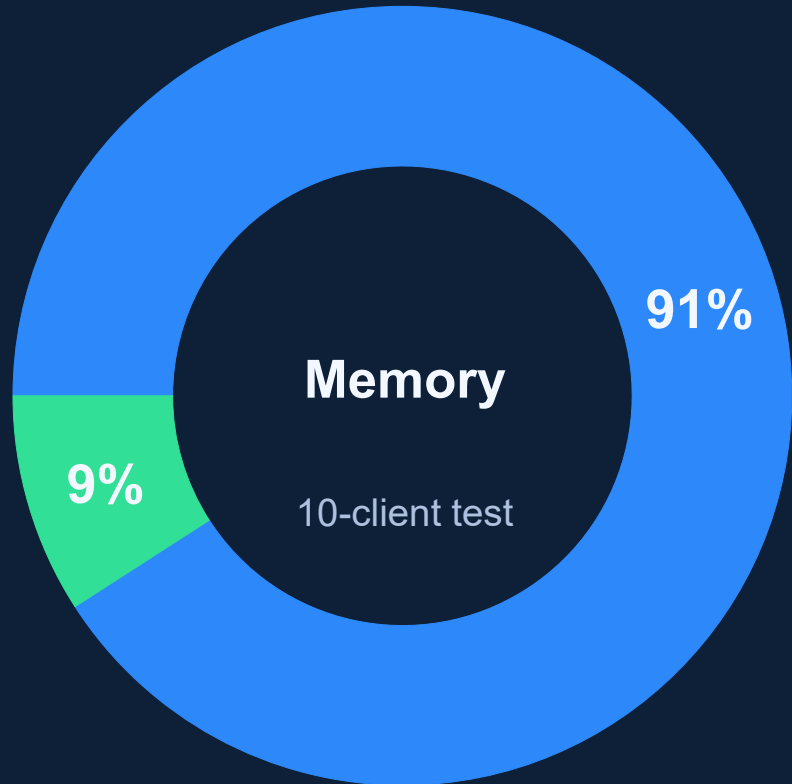
GitHub Copilot Runtime **Ported to Rust**

A faster, leaner foundation for agents



<https://aka.ms/copilot-migration>

Copilot runtime performance



■ Reduction ■ Remaining

91%

less memory

1,383 MB before / 126 MB after

65%

less processor time

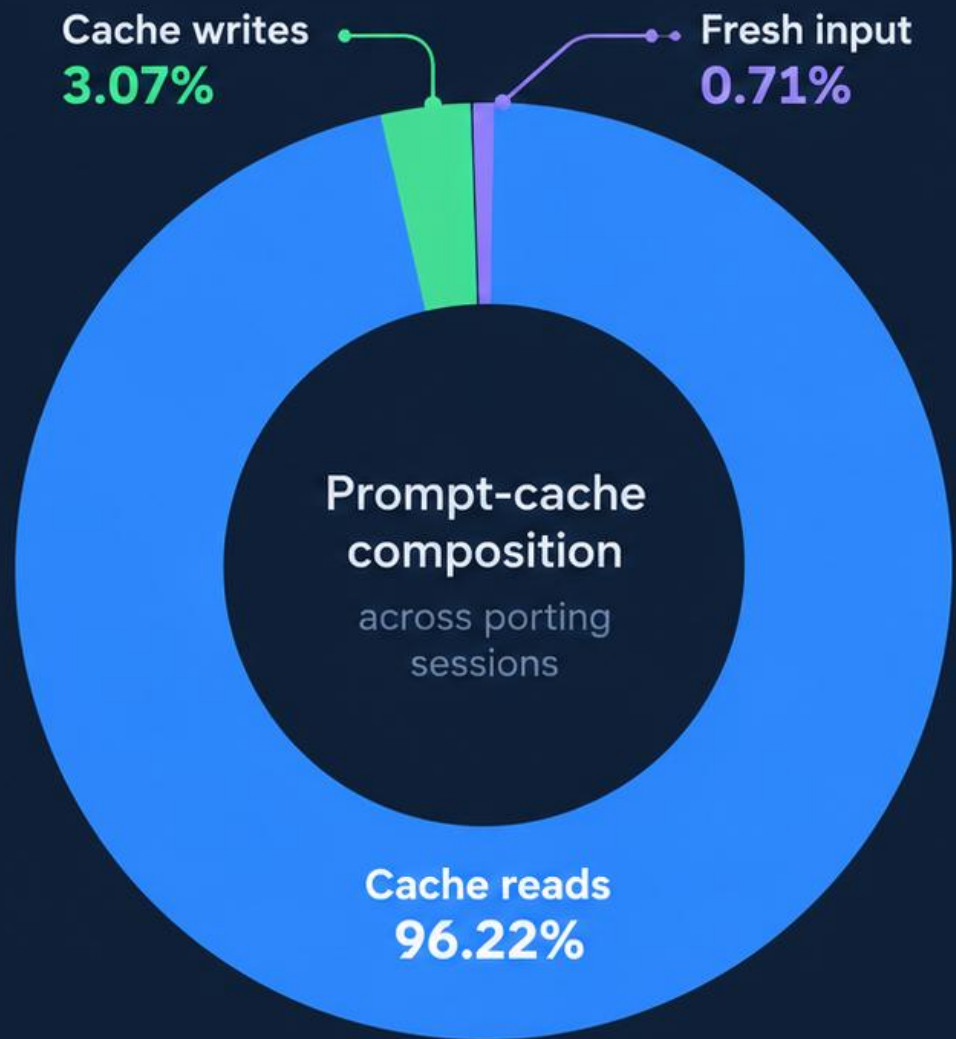
312 s before / ~110 s after

4×

faster session lifecycle

5.25 s before / 1.33 s after

Cost savings



\$120K

total spend

\$0.14

per line of production Rust

136.3B

total tokens

130.6B
cached reads

4.2B
cache writes

0.9B
fresh input

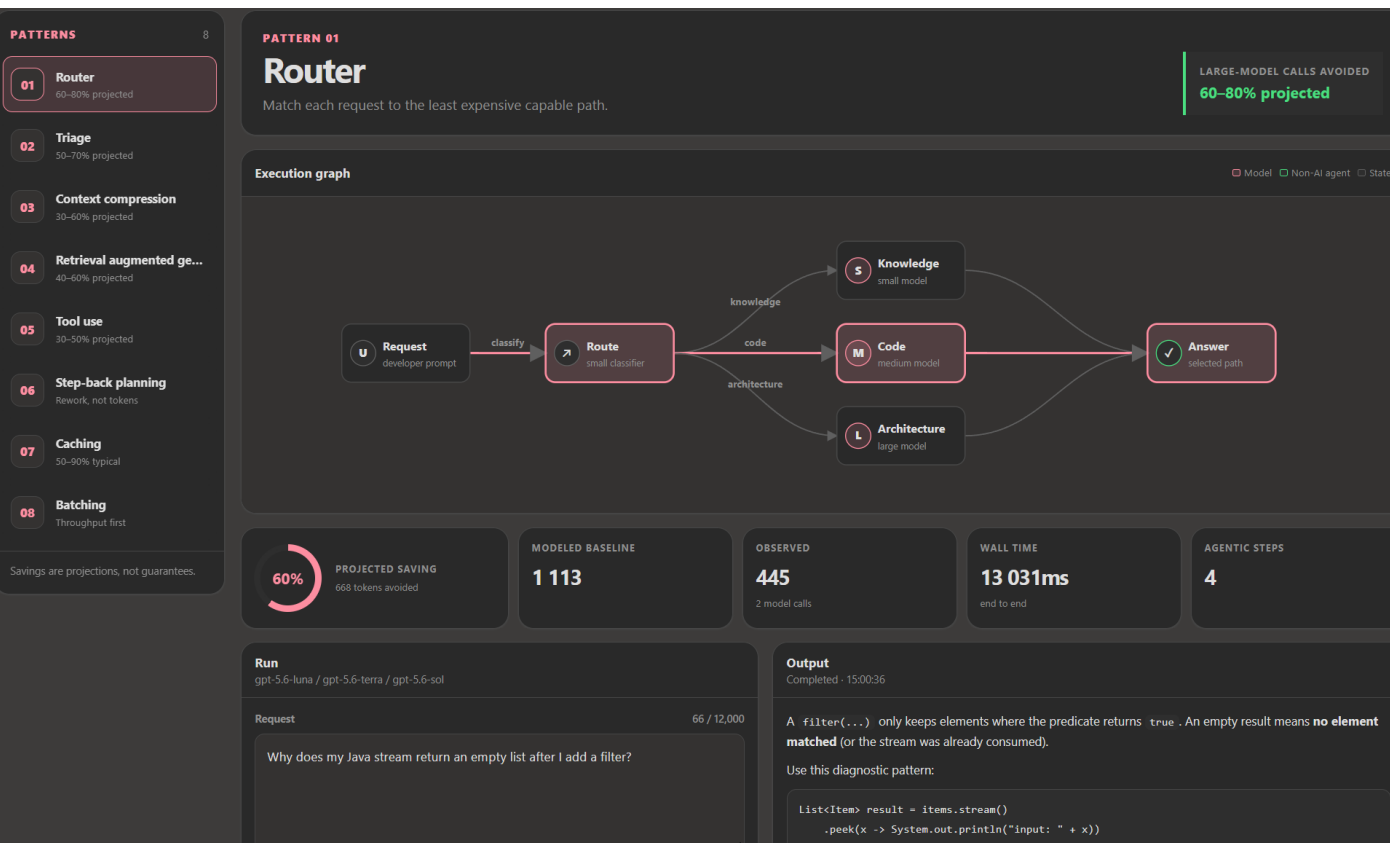
0.6B
output

📈 **96.22% prompt-cache hit rate** — primary savings lever

🔗 128 PRs

Agenda

- Introduction
- Demo 1 - 4 levers
- Demo 2 - Pricing
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- Resources



<https://aka.ms/costs>

MODEL



Router



Triage

PROMPT



Step-back
planning

CONTEXT



Compression



RAG



Tool use



Caching

OUTPUT



Batching

Models



GPT-5.6 Luna

Small / Fast



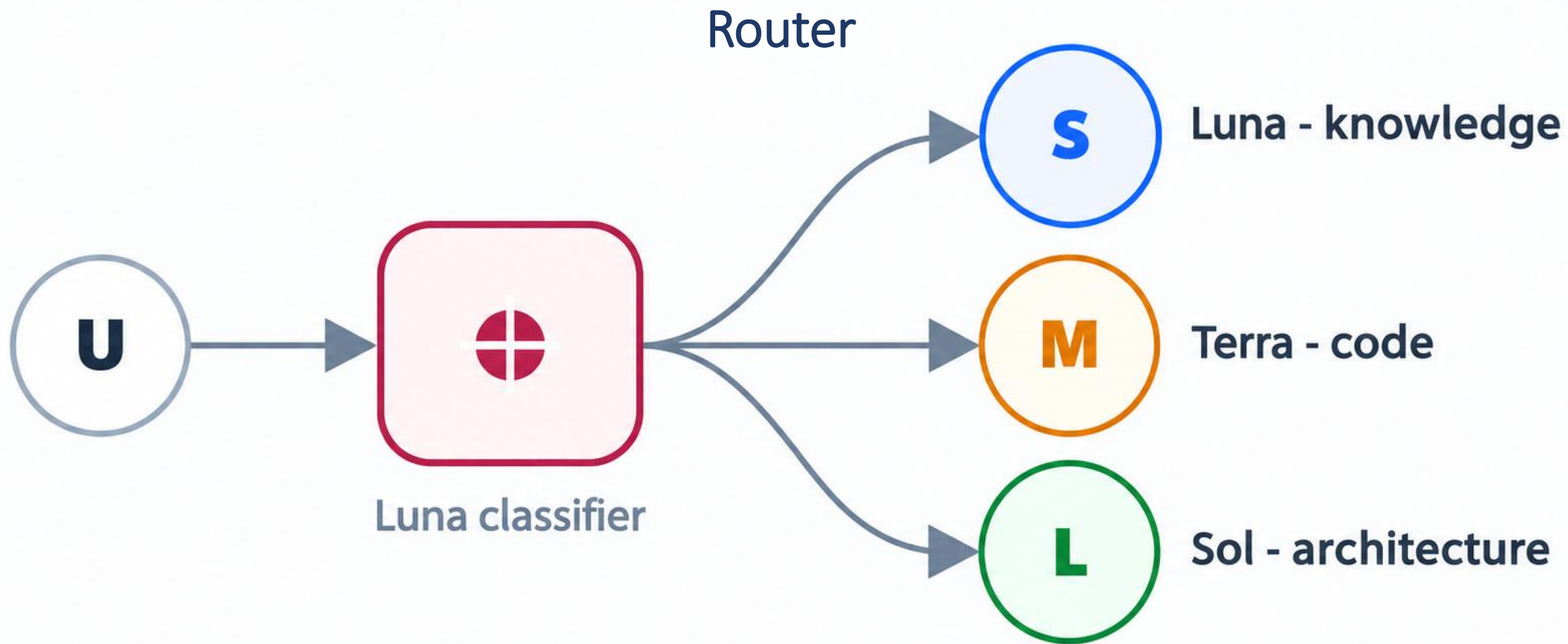
GPT-5.6 Terra

Medium

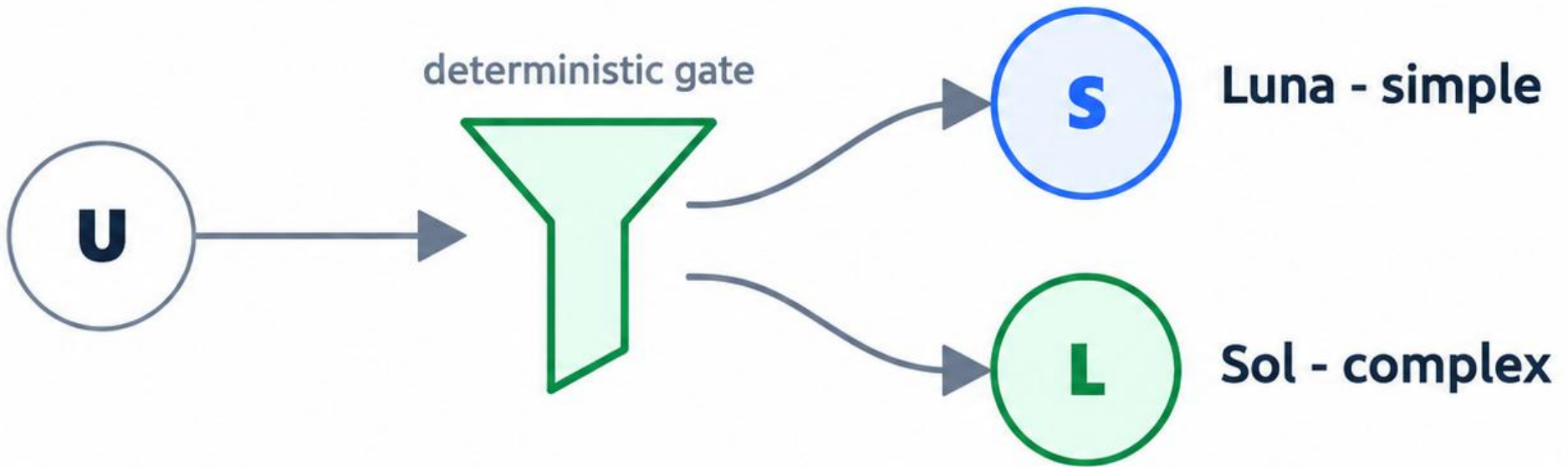


GPT-5.6 Sol

Large / Reasoning



Triage



MODEL



☒ Router

☒ Triage

PROMPT



☒ Step-back
planning

CONTEXT



☒ Compression

☒ RAG

☒ Tool use

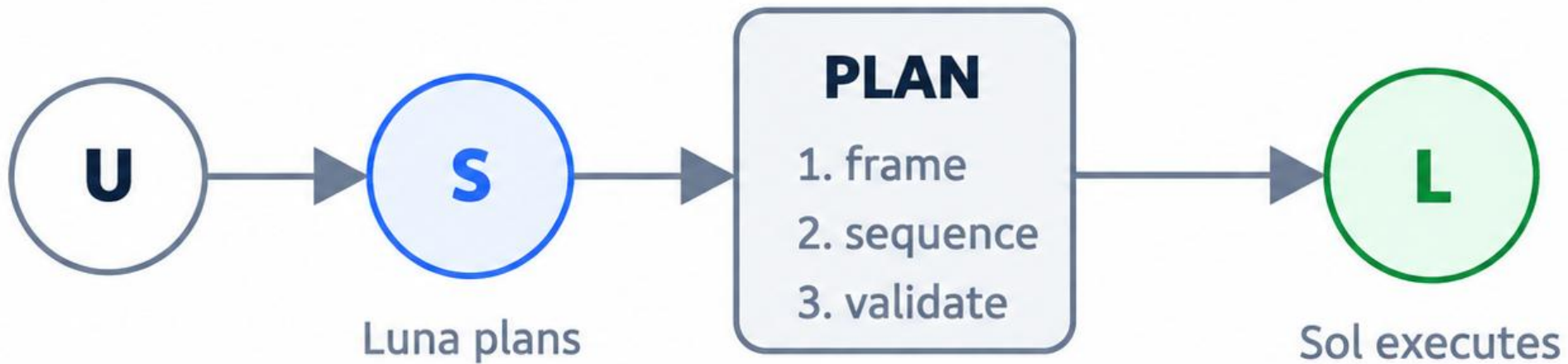
☒ Caching

OUTPUT



☒ Batching

Step-back planning



MODEL



☒ Router

☒ Triage

PROMPT



☒ Step-back
planning

CONTEXT



☒ Compression

☒ RAG

☒ Tool use

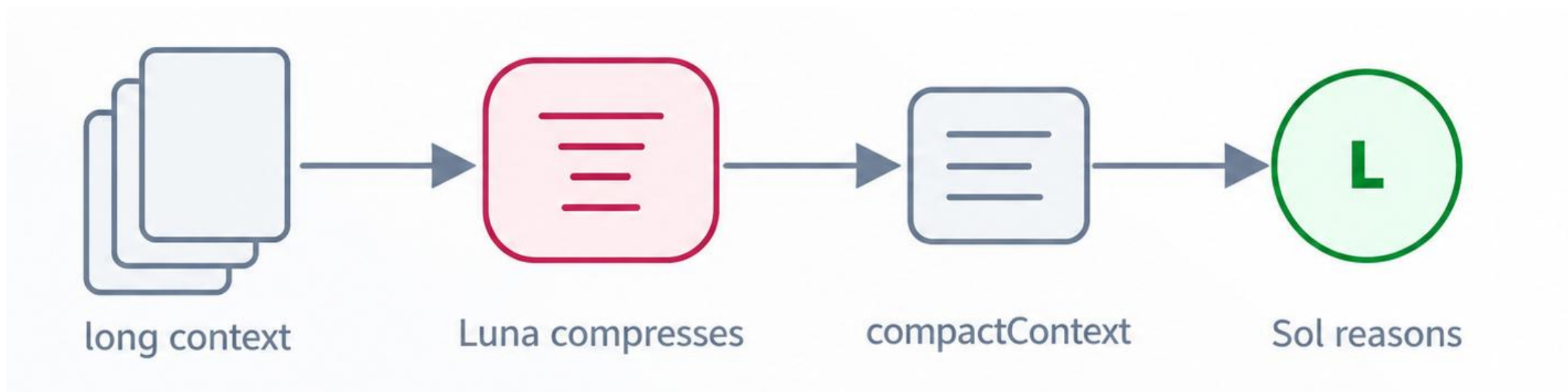
☒ Caching

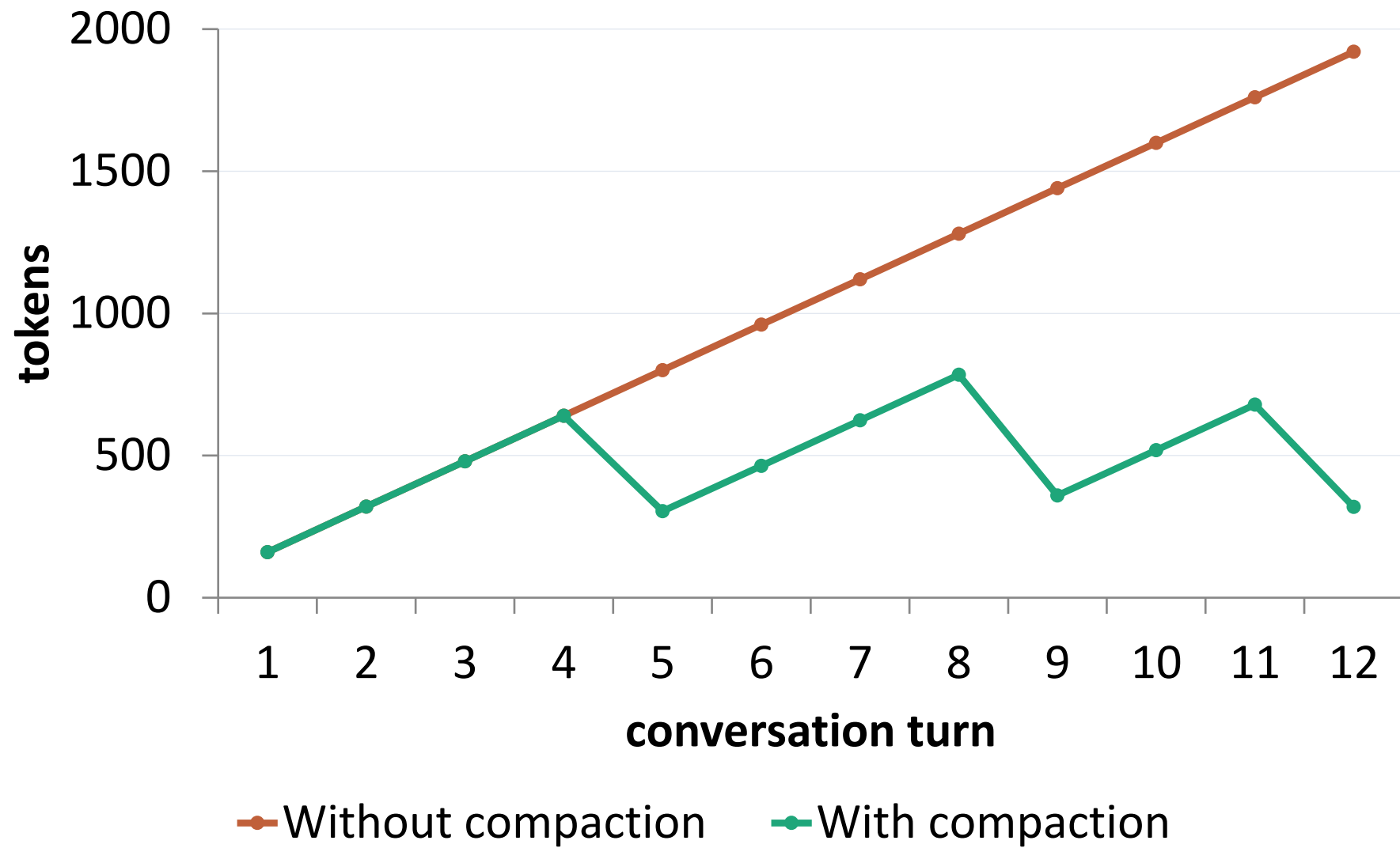
OUTPUT



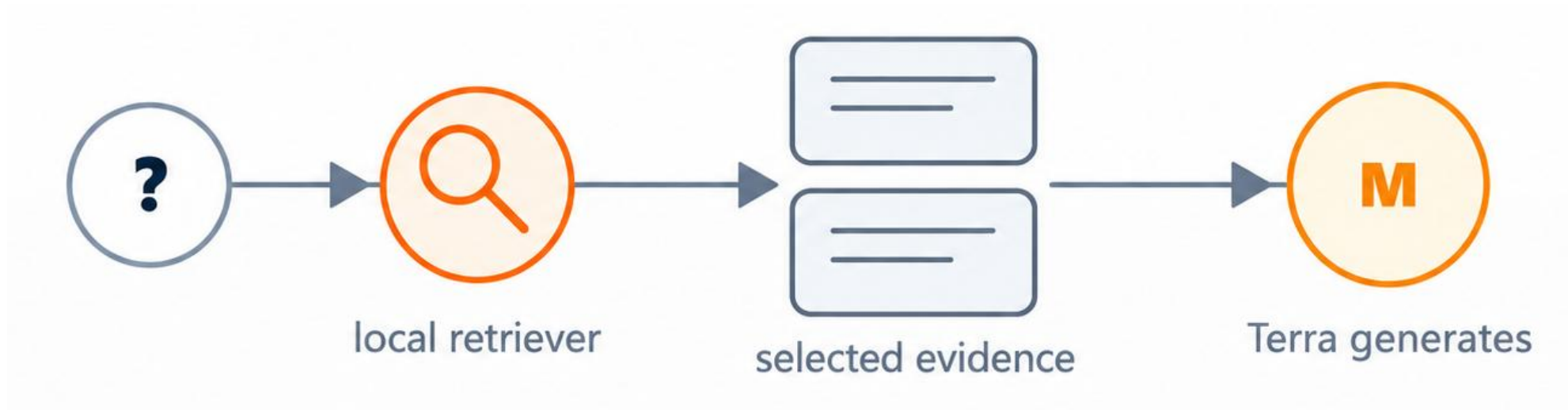
☒ Batching

Compression

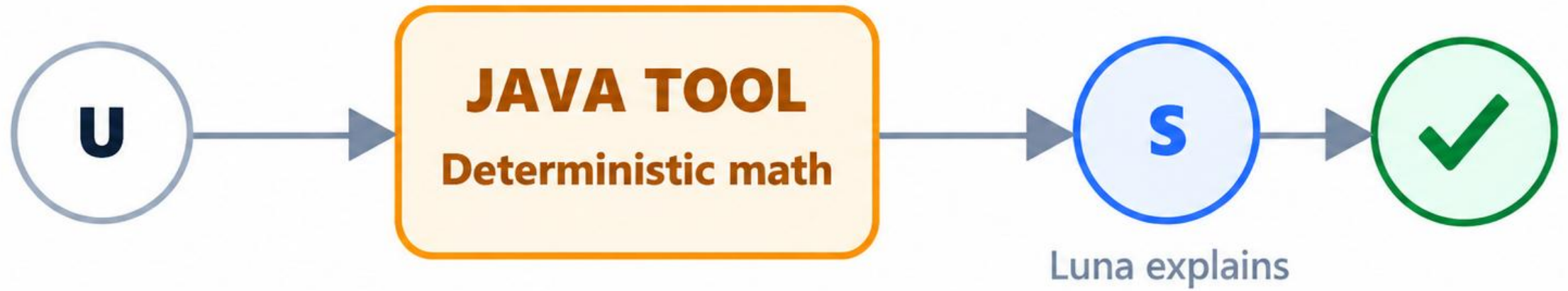




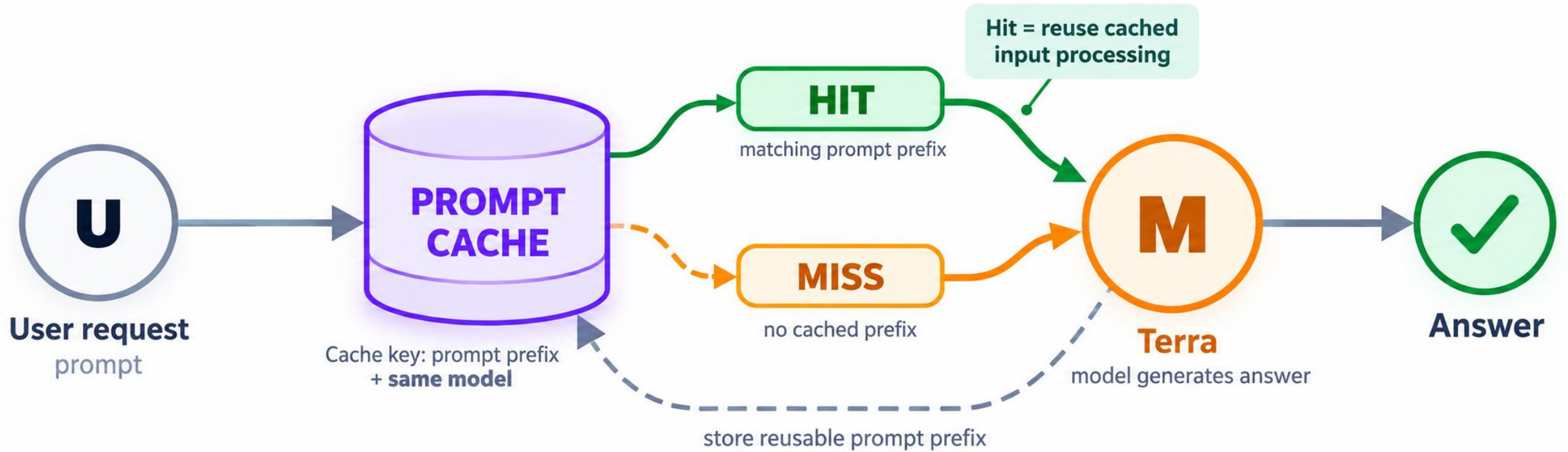
Retrieval-Augmented Generation (RAG)



Tools



Caching



MODEL



☒ Router

☒ Triage

PROMPT



☒ Step-back
planning

CONTEXT



☒ Compression

☒ RAG

☒ Tool use

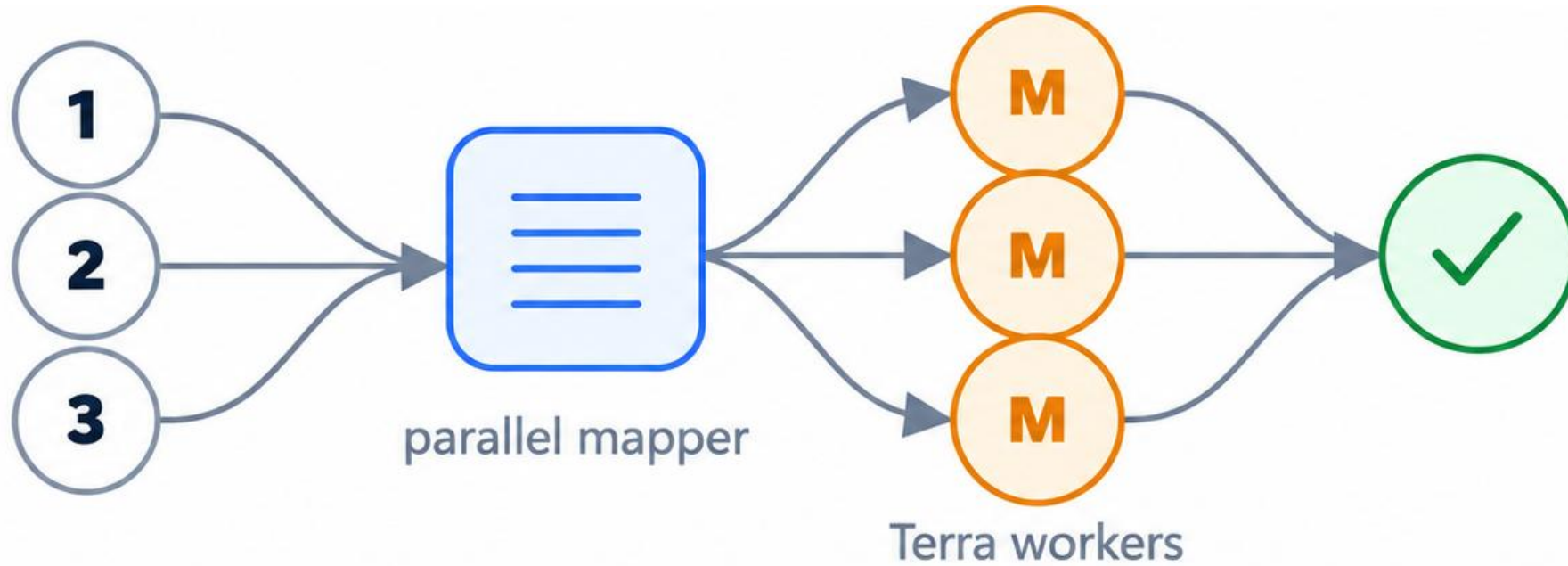
☒ Caching

OUTPUT



☒ Batching

Batching



MODEL



☒ Router

☒ Triage

PROMPT



☒ Step-back
planning

CONTEXT



☒ Compression

☒ RAG

☒ Tool use

☒ Caching

OUTPUT



☒ Batching

Agenda

- Introduction
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= 4 tokens

Your sentence

Seven shapes make words into falling blocks.

8 tokens / o200k_base

44/500



Seven



_shapes



_make



_words



_into



_falling



_blocks



.



Join game

```
{
  "BillingCurrency": "USD",
  "CustomerEntityId": "Default",
  "CustomerEntityType": "Retail",
  "Items": [
    {
      "currencyCode": "USD",
      "tierMinimumUnits": 0,
      "retailPrice": 1.2,
      "unitPrice": 1.2,
      "armRegionName": "eastus",
      "location": "US East",
      "effectiveStartDate": "2026-08-01T00:00:00Z",
      "meterId": "c6713056-7cef-508c-9e2e-7035984a4f44",
      "meterName": "5.6 luna ShortCo Opt Std G1 1M Tokens",
      "productId": "DZH318Z0T9WD",
      "skuId": "DZH318Z0T9WD/1166",
      "productName": "Azure OpenAI GPT5",
      "skuName": "5.6 luna ShortCo Opt Std G1",
      "serviceName": "Foundry Models",
      "serviceId": "DZH31600VF1F",
      "serviceFamily": "AI + Machine Learning",
      "unitOfMeasure": "1M",
      "type": "Consumption",
      "isPrimaryMeterRegion": true,
      "armSkuName": "56 luna ShortCo Opt Std G1"
    }
  ],
  "NextPageLink": null,
  "Count": 1
}
```

Currency: billing currency.

Retail price: list price charged.

Unit price: price per billing unit.

Region code: Azure region identifier.

Region name: human-friendly region name.

Effective date: when this price became active.

Meter name: full billing meter description.

Product name: product family.

SKU name: specific model/tier variant.

Service name: Azure service category.

Unit of measure: billed per 1M tokens.

Type: consumption-based billing.

TETRIS + LUNA

7. Tools

1. Pause / Restart

2. Score + points

3. Hold

4. Board

6. Controls

5. Next

11. Room

12. Rankings

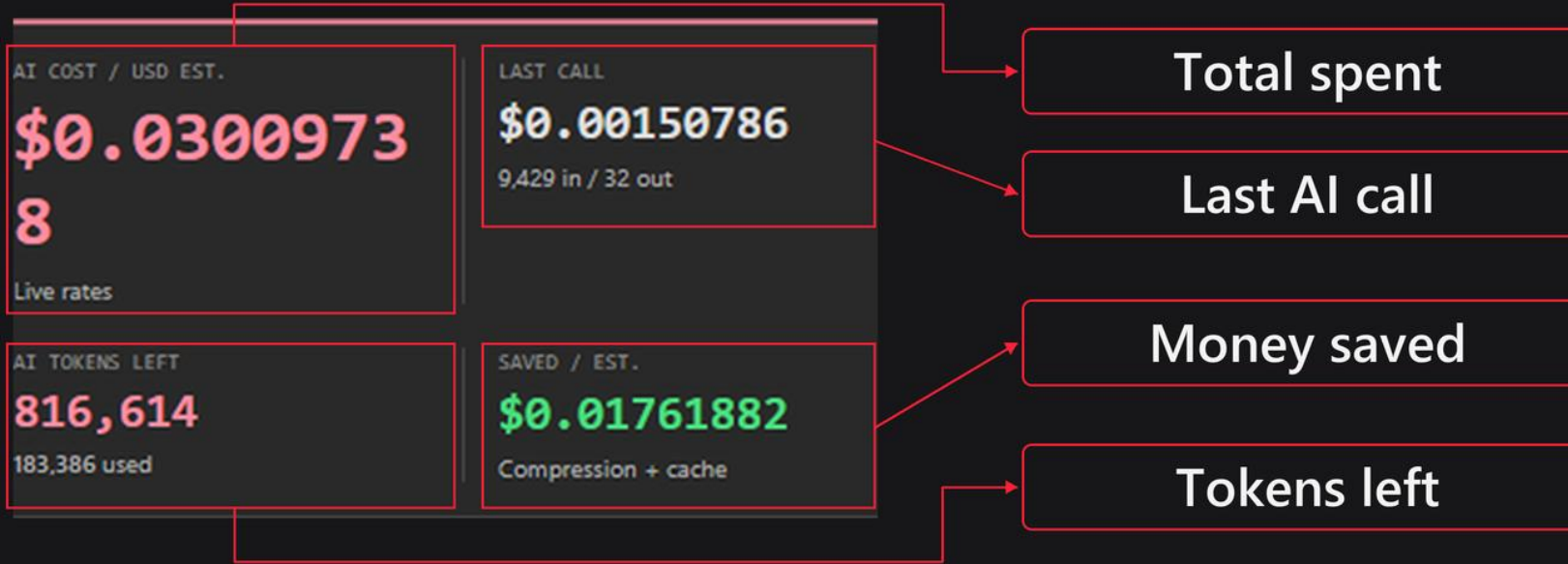
8. AI cost

9. AI play

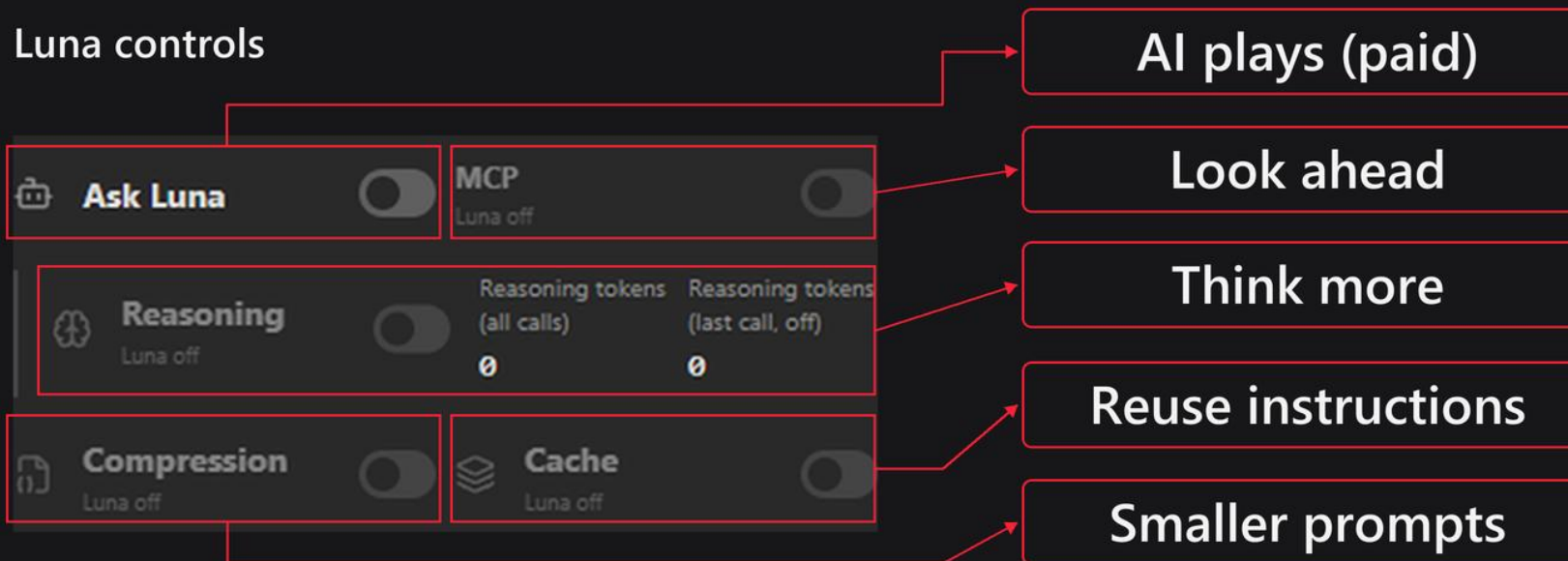
10. Status

AI COSTS + LUNA CONTROLS

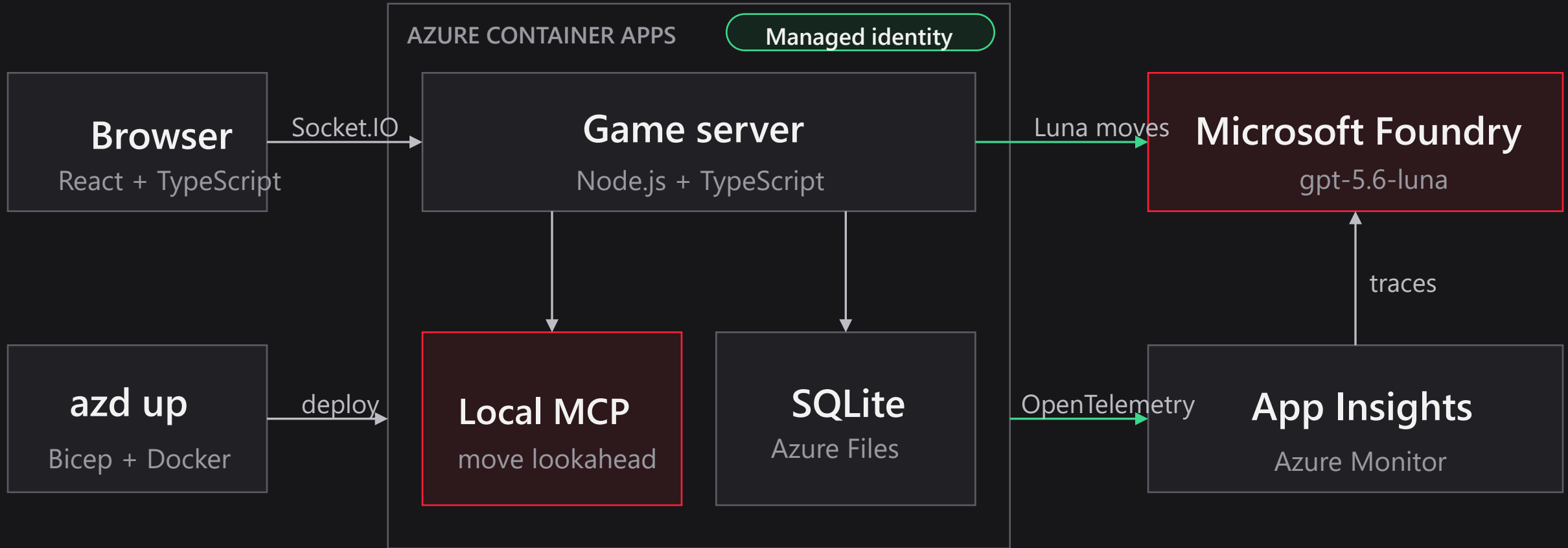
AI costs (USD estimates)



Luna controls



Architecture



SCORE
1,294 +38 Hard drop 19 rows x 2

LINES 6 LEVEL 1

HOLD

01 10

NEXT

make

my

blocks

Information & tools

- MCP results
- Cache activity
- Compression
- AI costs
- Leaderboard
- Share game
- Change sentence

AI COST / USD EST.
\$0.2008428

LAST CALL
\$0.0020656
9,266 in / 552 out

Live rates

AI TOKENS LEFT
11,421
988,579 used

SAVED / EST.
\$0.0104096
Compression + cache

Ask Luna

MCP
2-piece next

Reasoning
Low effort next

Reasoning tokens (all calls)
3,864

Reasoning tokens (last call, low)
512

Compression
Packed rows next

Cache
Reuse rules next

10 x 20 LVL 01



<https://aka.ms/tokenfall>

Agenda

- Introduction
- Demo 1 - 4 levers
- Demo 2 - Pricing
- Tracing and Scaling
- Resources

Tracing

tokenfall-luna

Traces [Preview](#) Insights [Preview](#) Monitor [Preview](#) Evaluation [Preview](#)

Search traces

Duration Status Tokens (In) Tokens (Out)

Create dataset

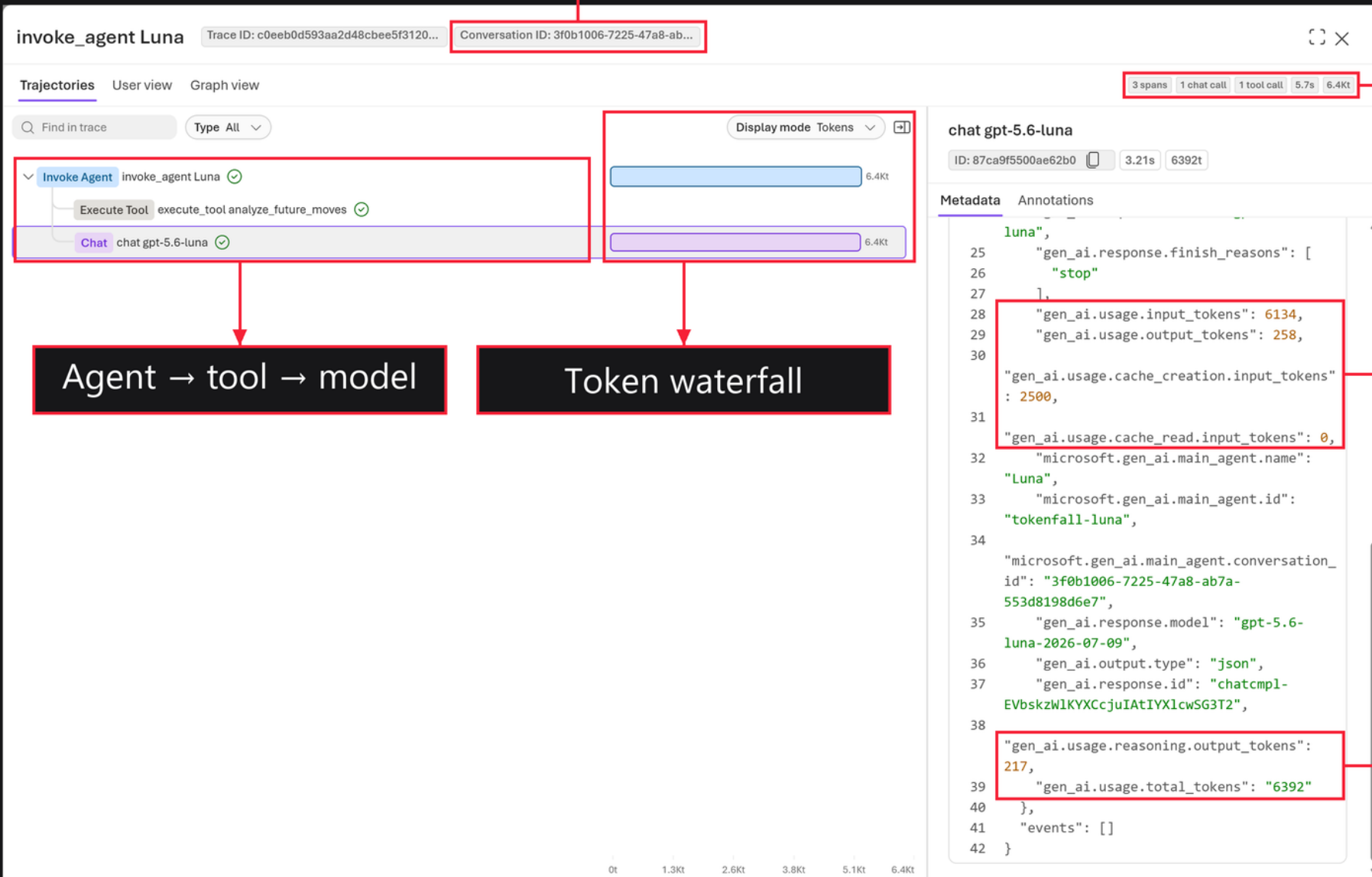
Trace ID	Status	Created at	Duration (s)	Tokens (In)	Tokens (Out)
1c466dde6...	✓ Completed	10/5/26, 2:15:07 PM	1.611	7,770	31
c0eeb0d59...	✓ Completed	10/5/26, 2:14:51 PM	5.720	6,134	258
6c09fa794...	✓ Completed	10/5/26, 2:14:40 PM	1.995	5,173	35

Linked Luna agent

Tokens in / out

One trace per move

One Luna move: MCP lookahead + reasoning



Game run

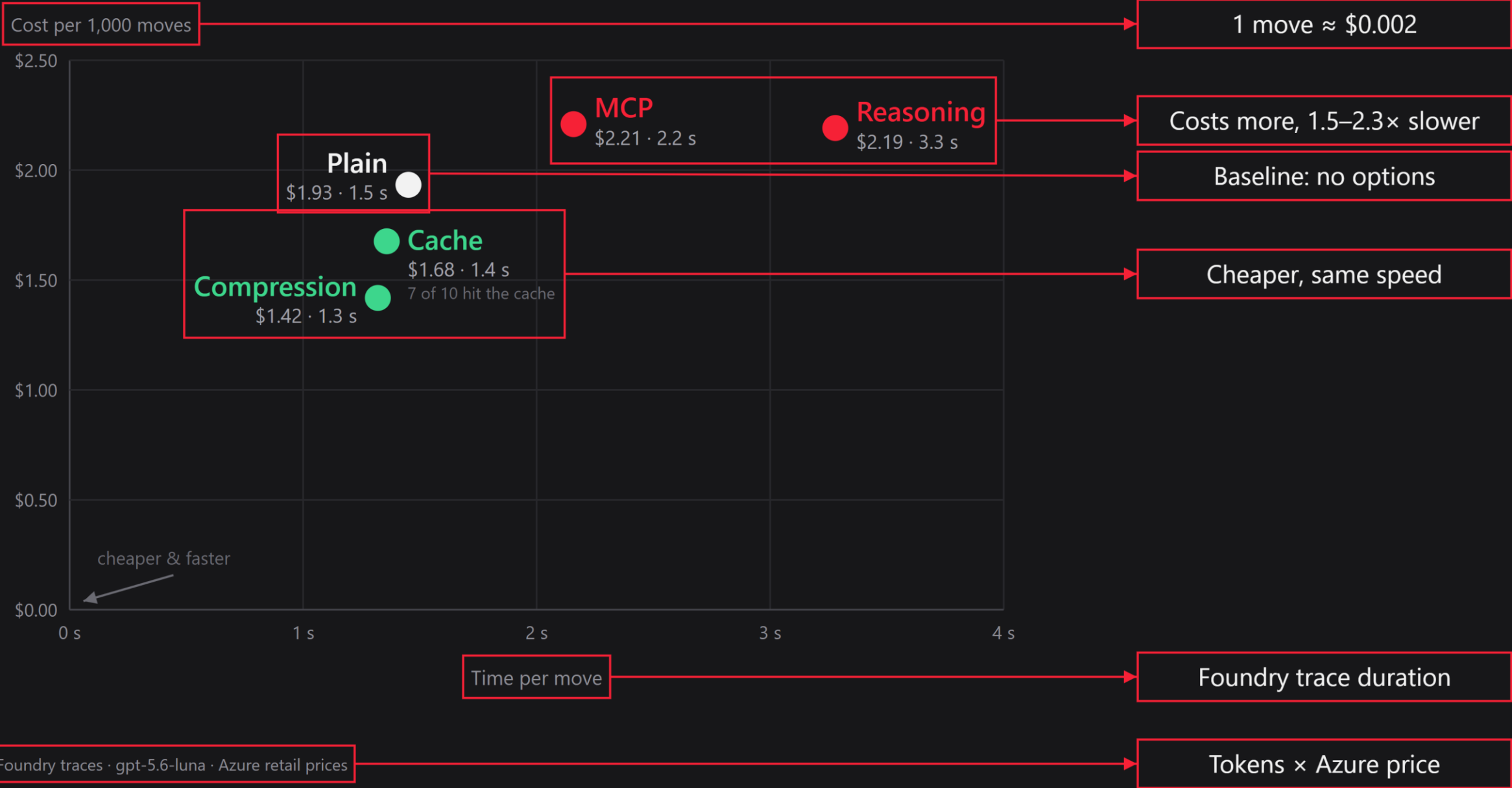
Trace totals

Input, output, cache

Reasoning + total

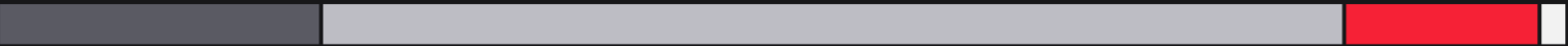
LUNA COST VS SPEED

Same 10 board positions for every option · game server with 2 CPUs



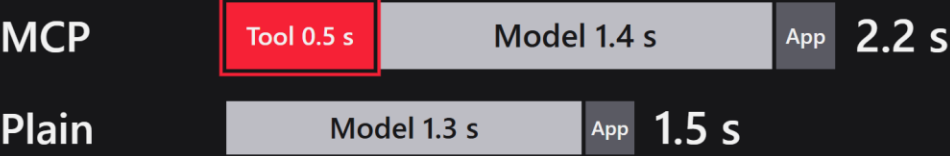
WHERE MCP'S \$2.21 GOES

Cost of 1,000 Luna moves with MCP, averaged from 10 Foundry traces



	TOKENS PER MOVE	× PRICE PER 1M	= PER 1,000 MOVES	
■ Instructions Game rules and reply format	2,276	\$0.20	\$0.46	Same every move: cacheable
■ Game state Board, pieces and every legal move	7,200	\$0.20	\$1.44	Most of the cost: 65%
■ MCP lookahead Tool results for the next two moves	1,375	\$0.20	\$0.27	What MCP adds: +14%
■ Luna's reply The chosen move and a short tip	33	\$1.20	\$0.04	Output: 6× price per token
Total	10,884		\$2.21	= \$0.0022 per move
				Tool: +0.5 s, no AI call was 2.1 s on 0.5 CPU

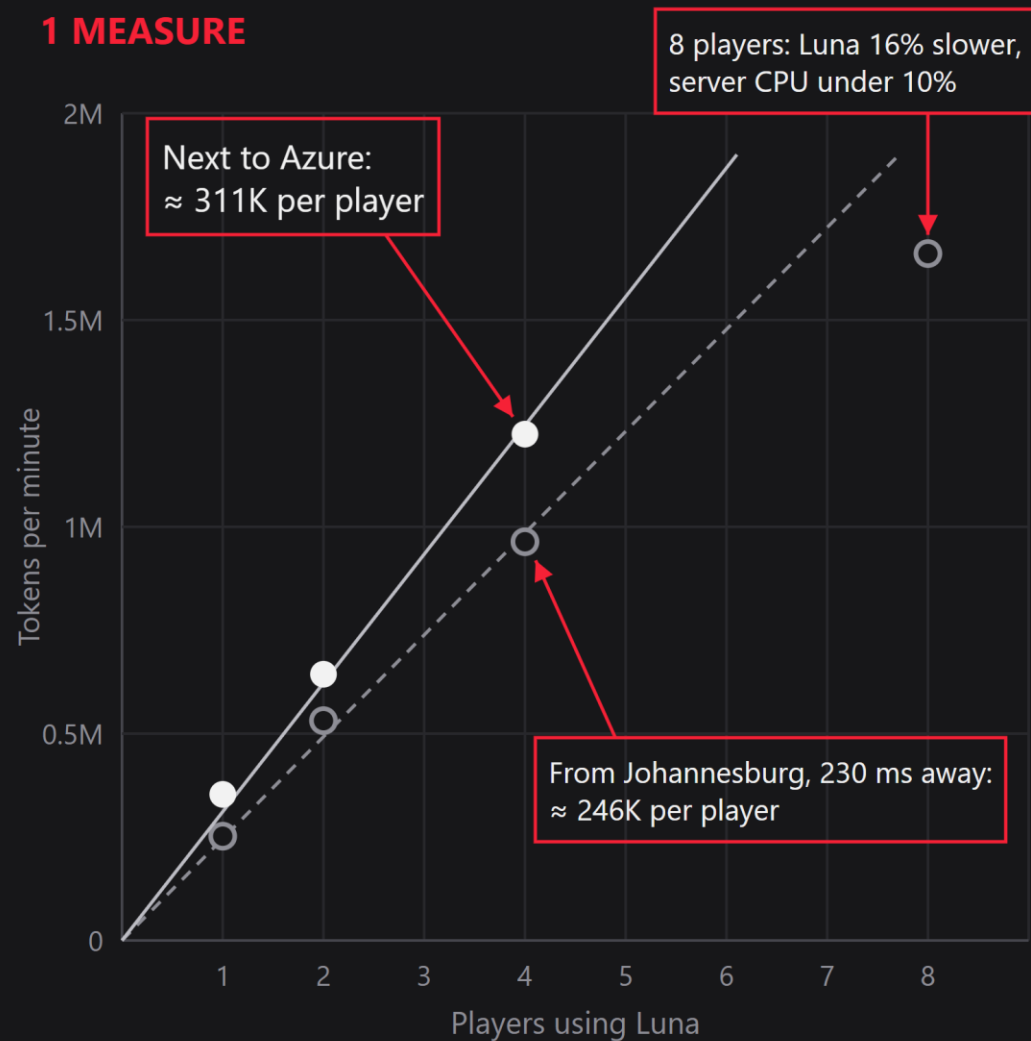
TIME PER MOVE



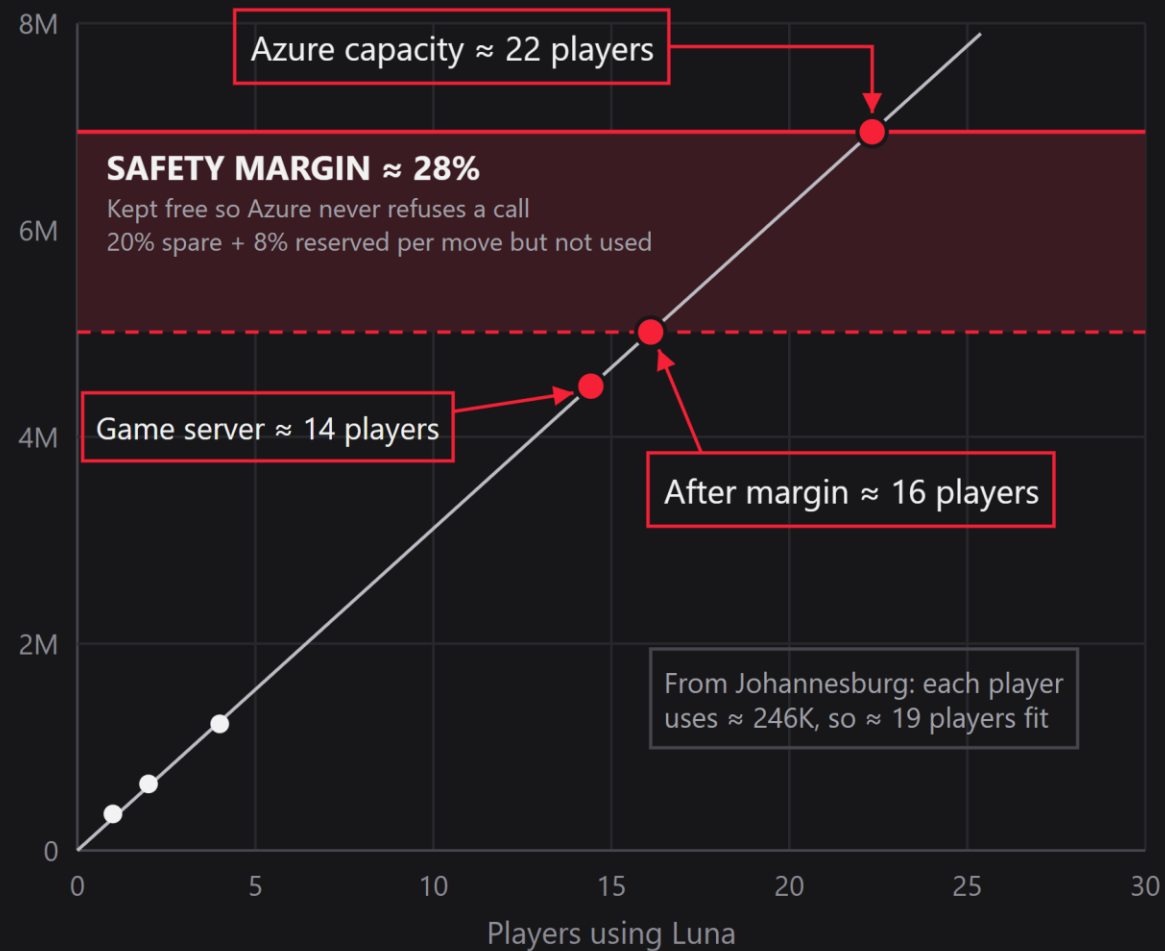
HOW MANY LUNA PLAYERS AT ONCE?

Plain Luna moves on the live game server: 1 replica, 2 CPUs, 4 GiB

1 MEASURE

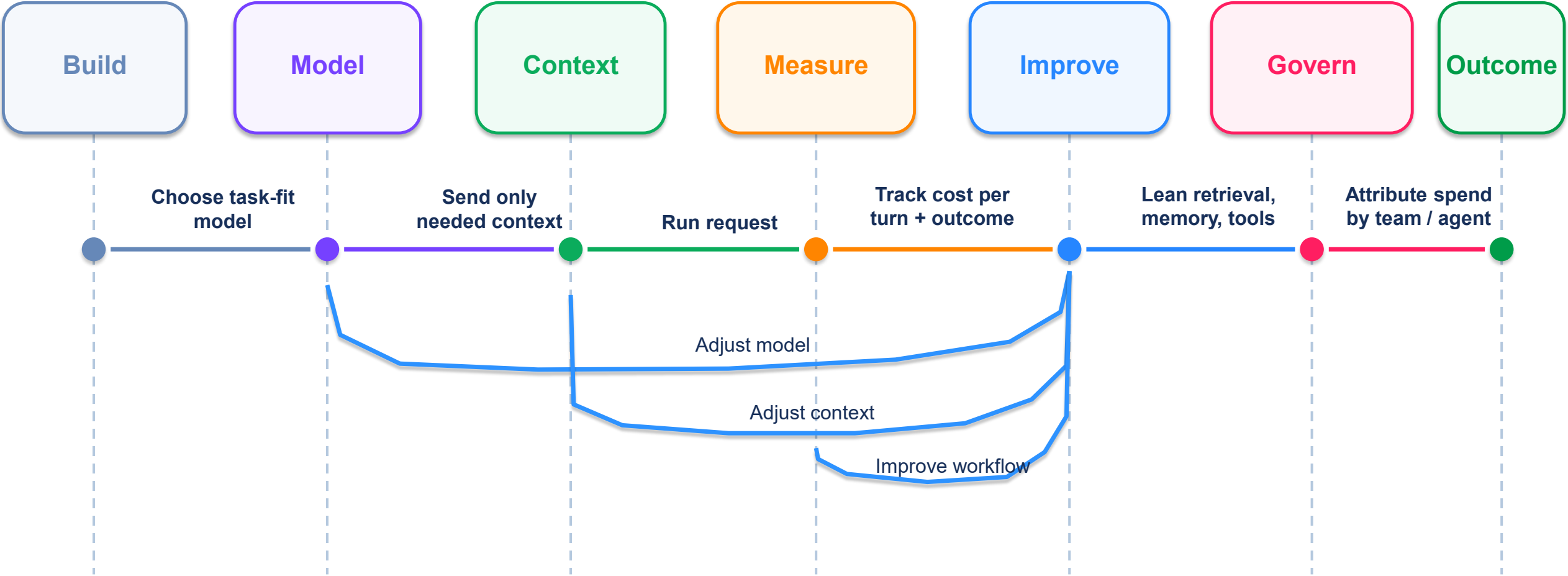


2 PREDICT



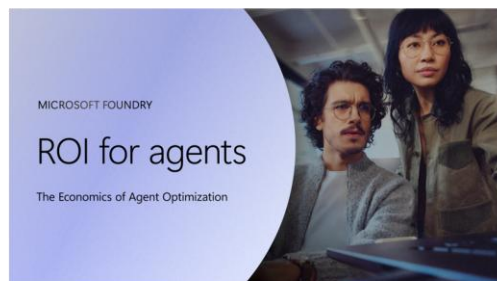
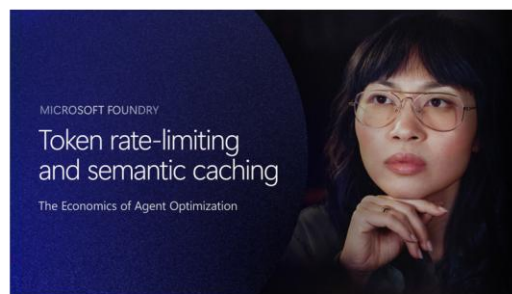
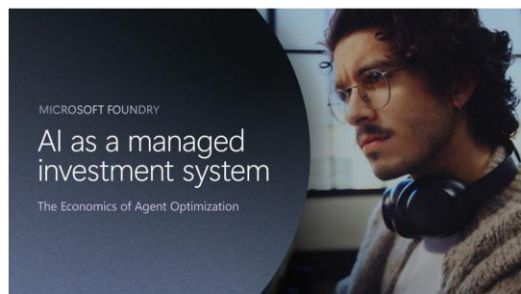
Plain moves, one minute per step · Next to Azure: bots inside the app's container · Johannesburg: bots on the internet, 230 ms away

Token Economics Checklist

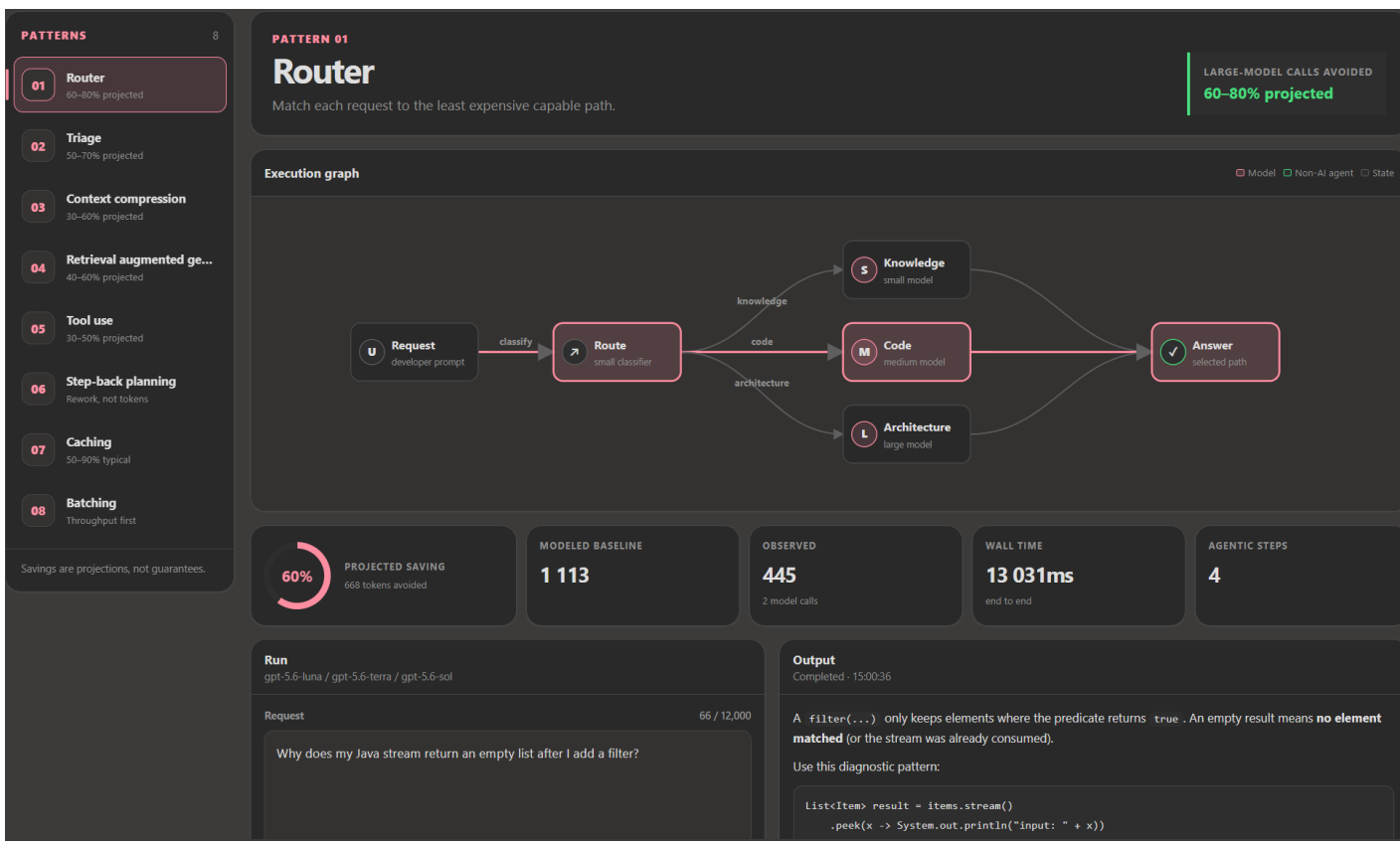


Agenda

- Introduction
- Demo 1 - 4 levers
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<https://aka.ms/token-economics>



<https://github.com/roryp/token-design-patterns>

SCORE
1,294 +38 Hard drop 19 rows x 2

LINES 6 LEVEL 1

HOLD

01 10

Tokens

!

make

my

blocks

10 x 20 LVL 01

Information & tools

- MCP results
- Cache activity
- Compression
- AI costs
- Leaderboard
- Share game
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AI COST / USD EST.
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Live rates

AI TOKENS LEFT
11,421
988,579 used

SAVED / EST.
\$0.0104096
Compression + cache

Ask Luna

MCP 2-piece next

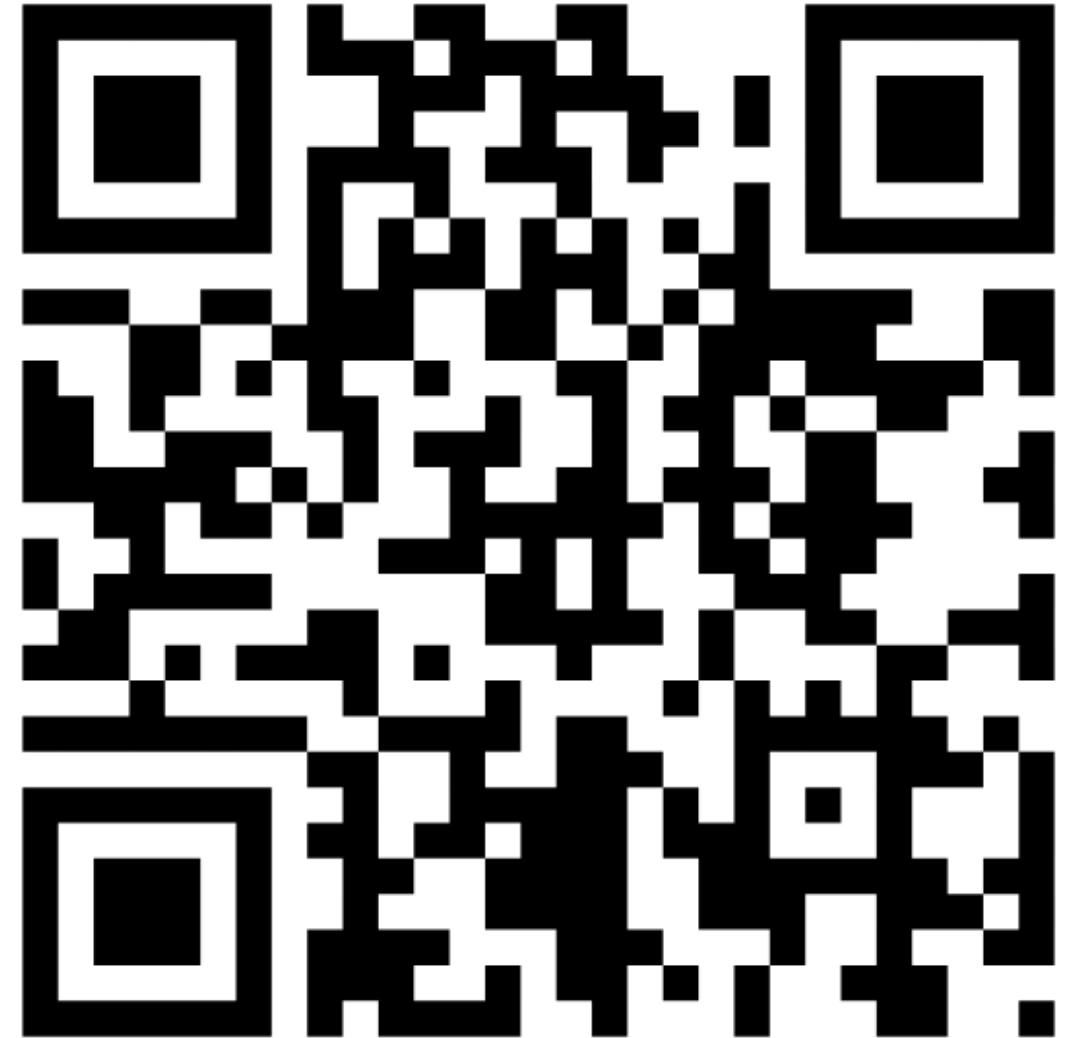
Reasoning Low effort next

Reasoning tokens (all calls) 3,864

Reasoning tokens (last call, low) 512

Compression Packed rows next

Cache Reuse rules next



<https://github.com/roryp/tokenfall>